

AU2026
THE DESIGN & MAKE CONFERENCE

AS2841: Autodesk Forma to Energy: AI-Powered Workflows for Carbon Insight

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Speakers



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Agenda

01

Current Bottlenecks
Forma Carbon Insights Vision

02

Forma Building Design → Forma Carbon Insights

03

Forma Carbon Insights → Customized HVAC Systems (concept demo)
Forma Carbon Insights → Customized SHW Systems

04

Revit → OS MCP

05

Conclusions and Q&A

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Current Bottleneck

**Energy and
carbon
analysis today
typically
happens **too
late****

Workflows between Forma, Revit,
and external analysis tools are
manual and **disconnected**.

Connected
Workflows

Specialized simulation skills are
scarce and **concentrated** in a
small number of energy modelers.

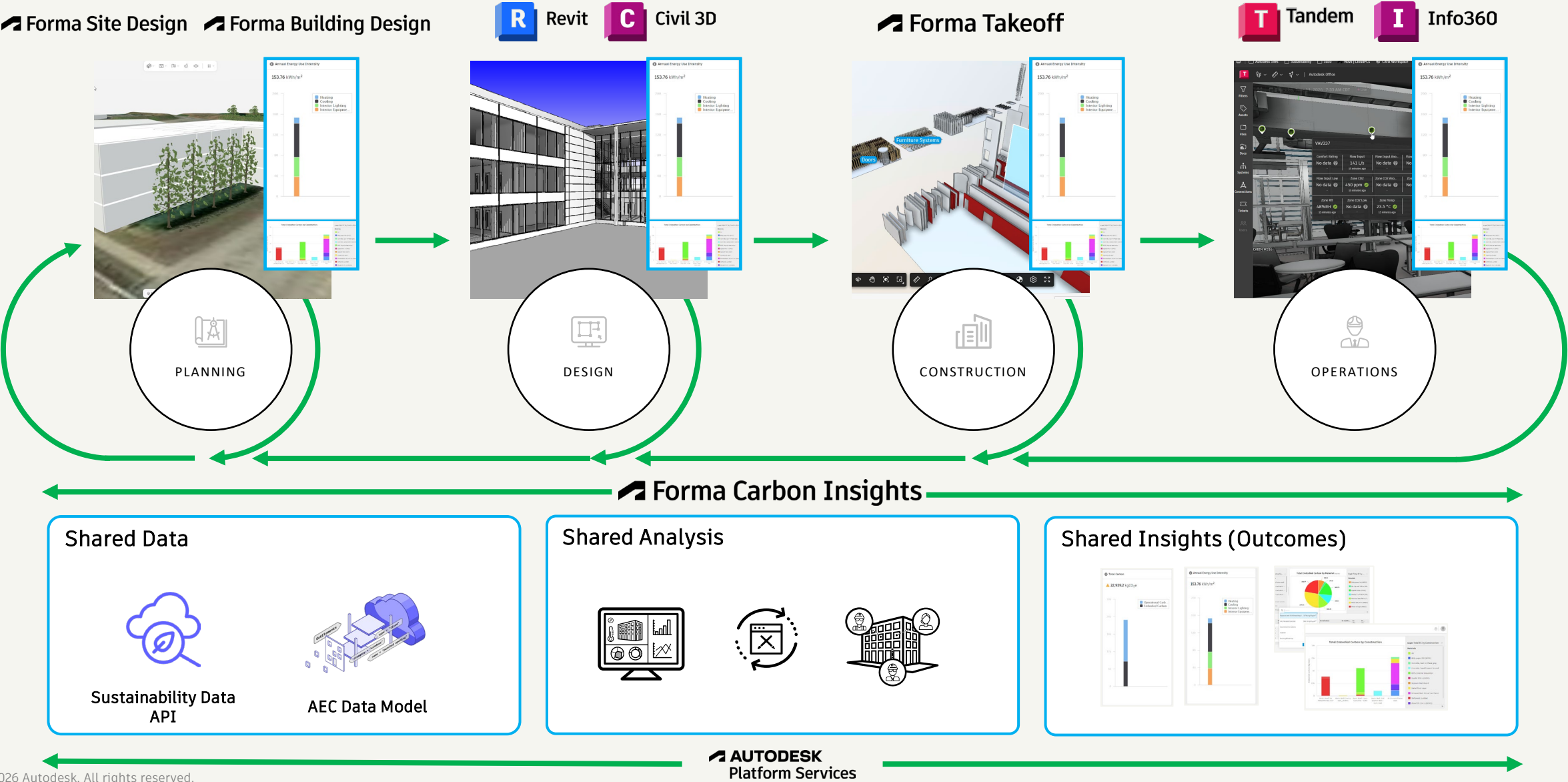
Leverage
Best
Practices

Analysis waits for a specialist to
become available — by which
point the geometry, systems, and
envelope are **already fixed**.

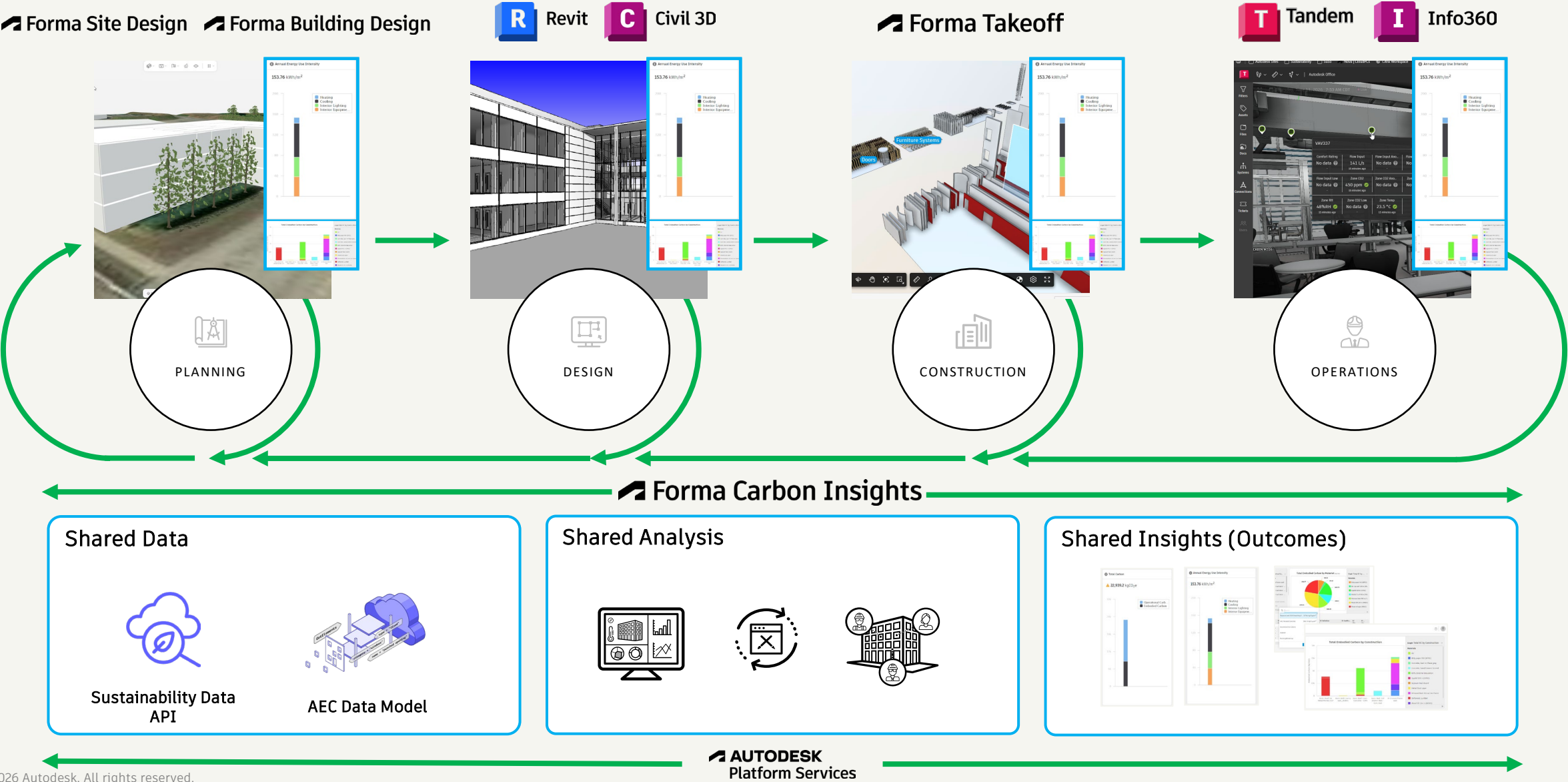
Deploy
Agentic
Tools

Solutions: Connect tools and put expert-level modeling capability into the hands of non-specialists.

Forma Carbon Insights (FCI) Vision



Forma Carbon Insights (FCI) – Available Today





Forma Building Design to Revit to Forma Carbon Insights

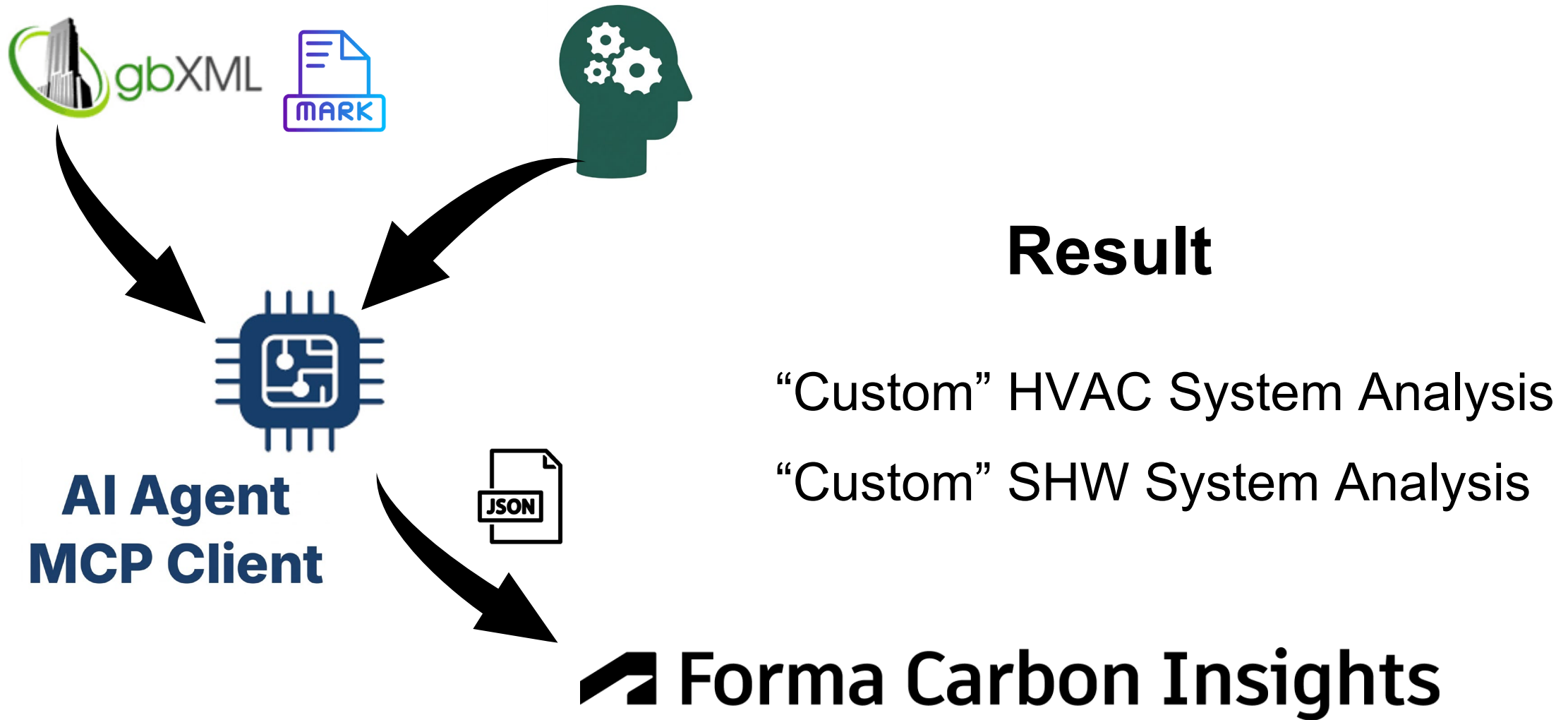
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Forma Carbon Insights – Custom HVAC and SHW Systems

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Concept Demo: Modeling Custom HVAC / Custom SHW

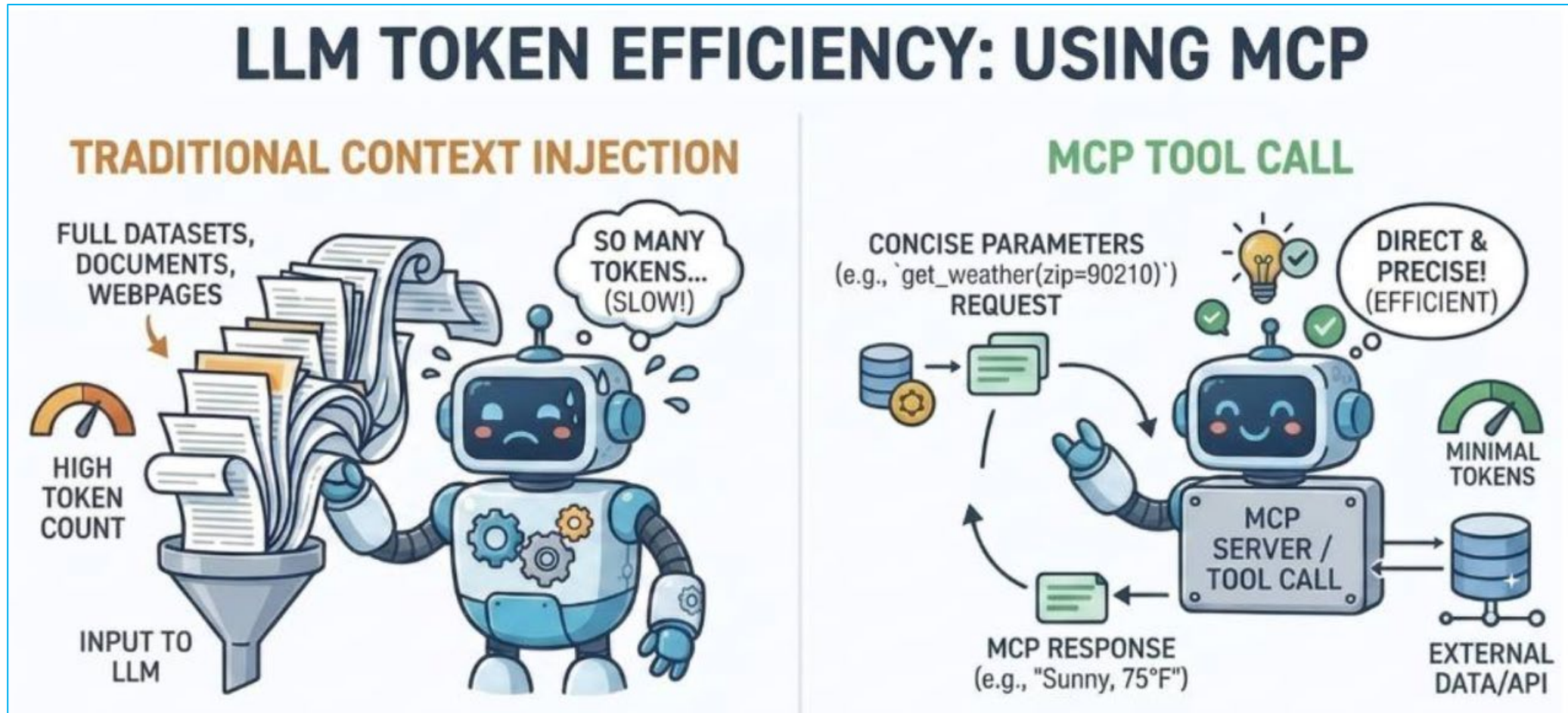




Revit to OpenStudio MCP

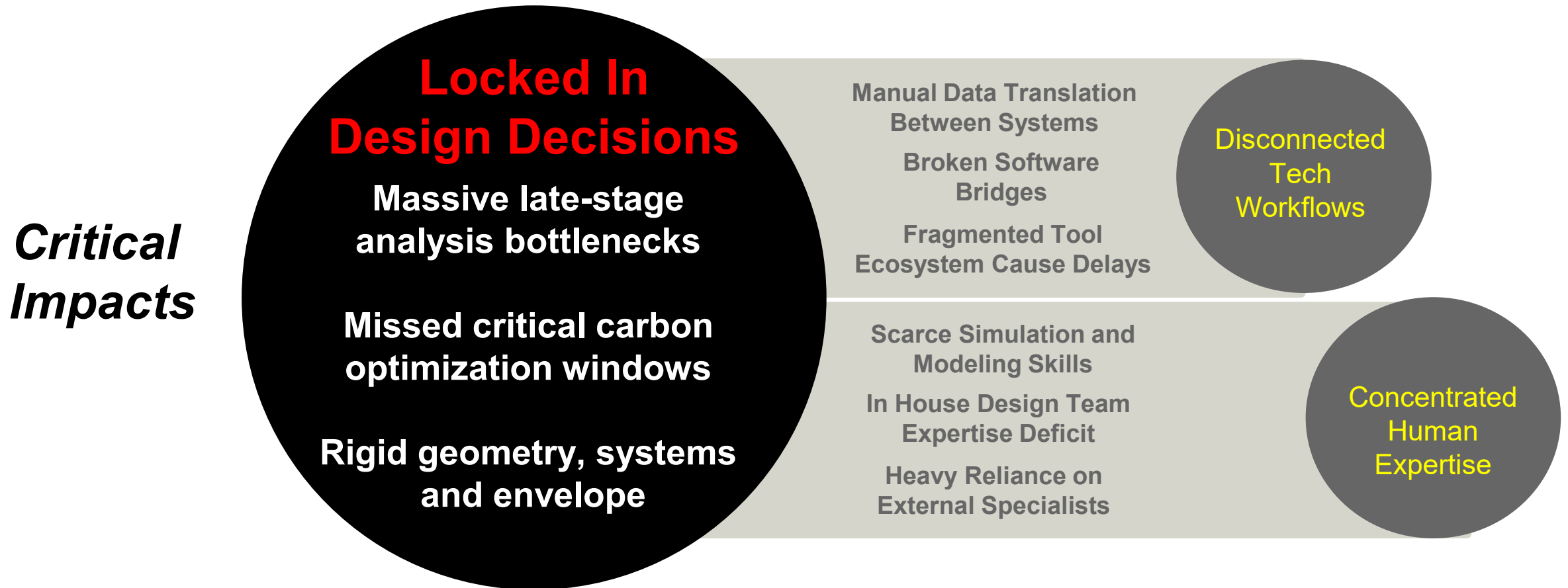
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Why use an MCP?



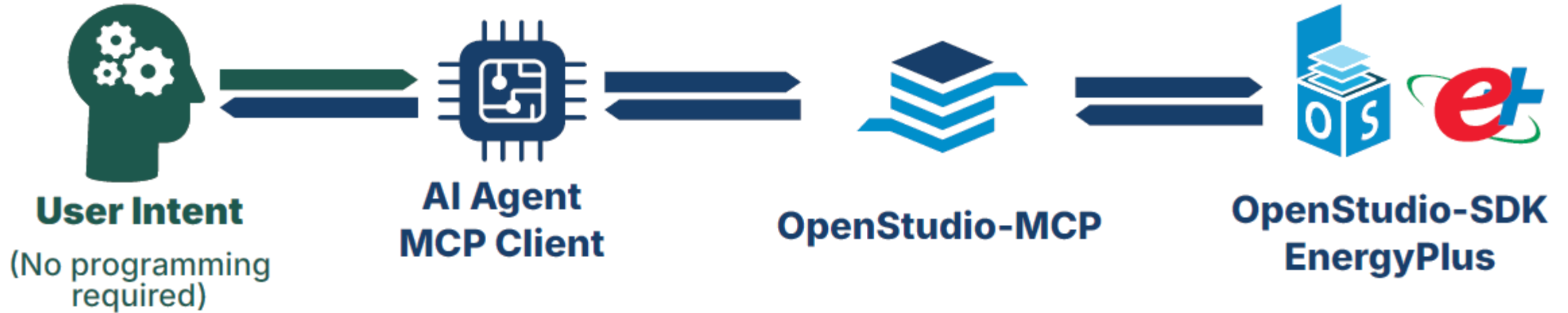
MCP workflows support **sandboxed** and **well-defined** data retrieval

Energy Analysis: Too Late to Influence Core Choices



OpenStudio MCP based analysis workflows can solve these problems

OpenStudio-MCP



An MCP server that lets AI agents create, modify, simulate, and analyze building energy models using MCP Tools.



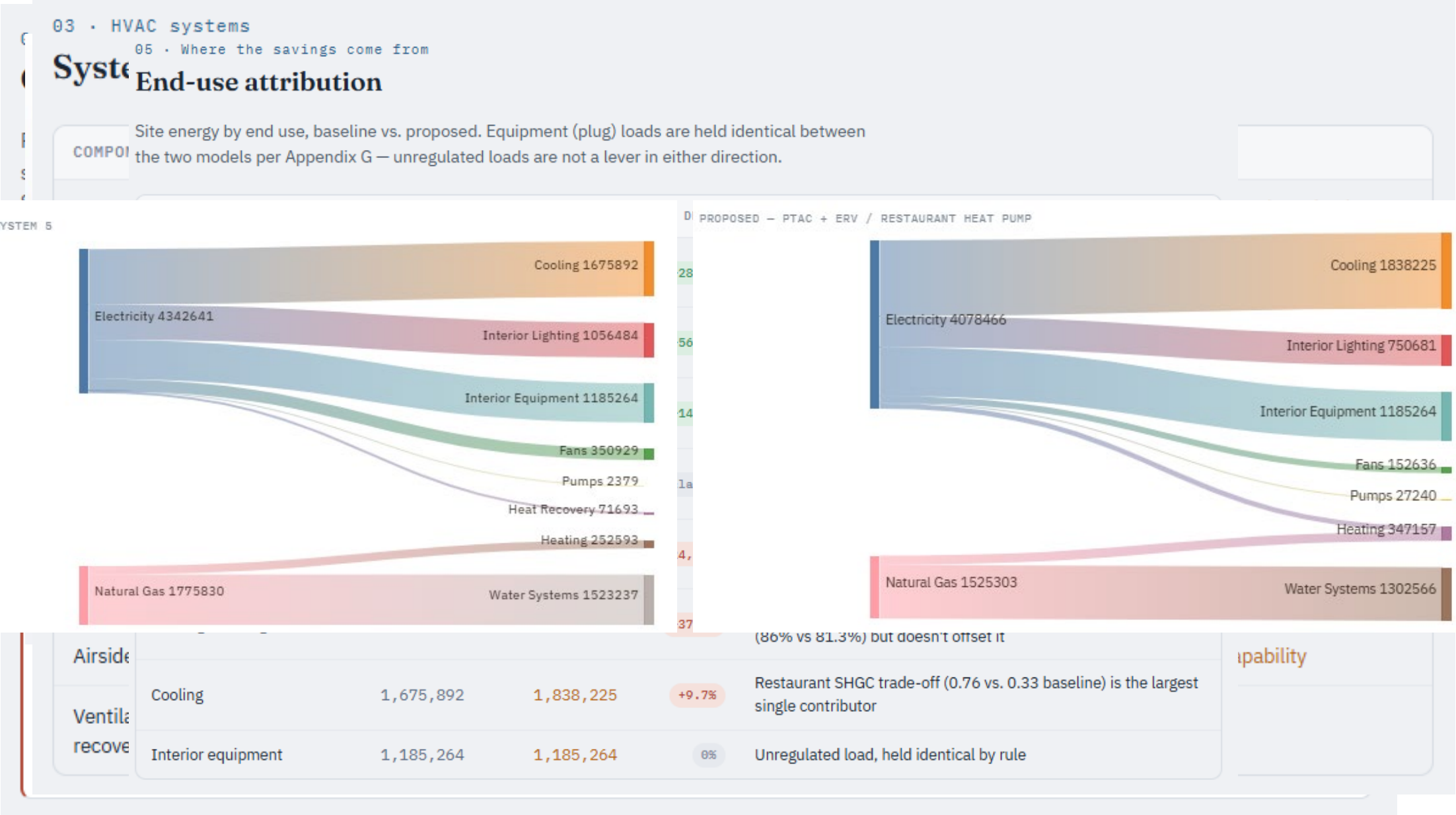
U.S. DEPARTMENT
of **ENERGY**



National
Laboratory
of the Rockies

[NatLabRockies/openstudio-mcp](https://github.com/NatLabRockies/openstudio-mcp)

OS-MCP: Report Generation



Why OS MCP



Matters...

01

Flexible Hosting

- *Secure: Uses Simulation Sandbox*
- *Install Local - Single User*
- *Install Remote - Multi-User*
- *Configurable upload and download options*

02

Knowledge Sharing

- *Organizations can leverage shared modeling capacity*
- *Organizational BEM knowledge can be harvested*
- *Users can be onboarded by issuing tokens rather than provisioning workstations / licenses.*

03

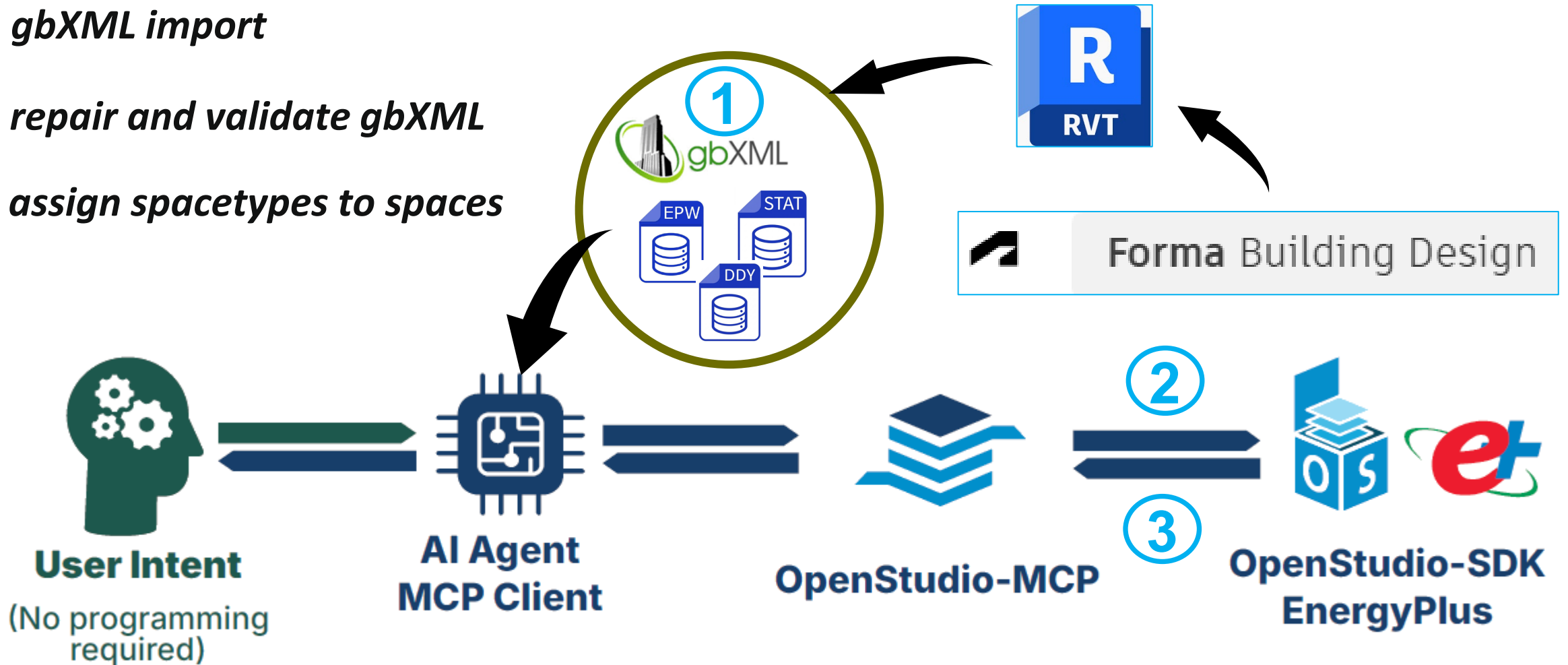
Disruptive BEM

- *Turns BEM from a per-seat desktop activity into shared infrastructure.*
- *Users can compare options, revise and rerun (while sharing).*

“putting expert-level modeling capability into the hands of non-specialists”

Three **New** OS MCP Tools:

- 1) *gbXML import*
- 2) *repair and validate gbXML*
- 3) *assign spacetypes to spaces*

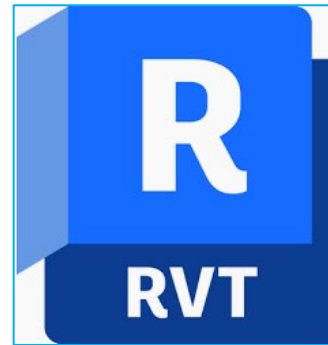


Tool: gbXML import: ①

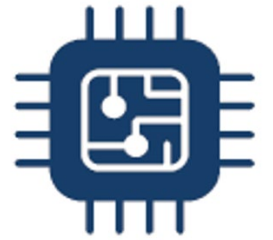
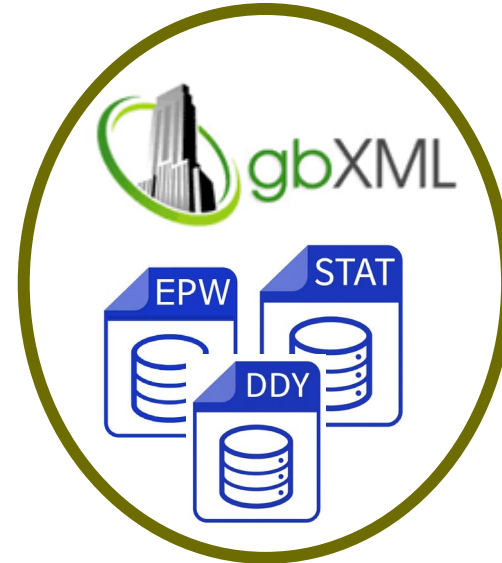
Prompt

I have a gbXML export from Revit at `/inputs/25_SpacesOneZE.xml`, along with its project weather file `/inputs/USA_MA_Boston-Logan.Intl.AP.725090_TMY3.epw` (and matching `.stat` / `.ddy`). Import it into OpenStudio and tell me what's in it.

- Capture
- Download
- Upload to OS-MCP



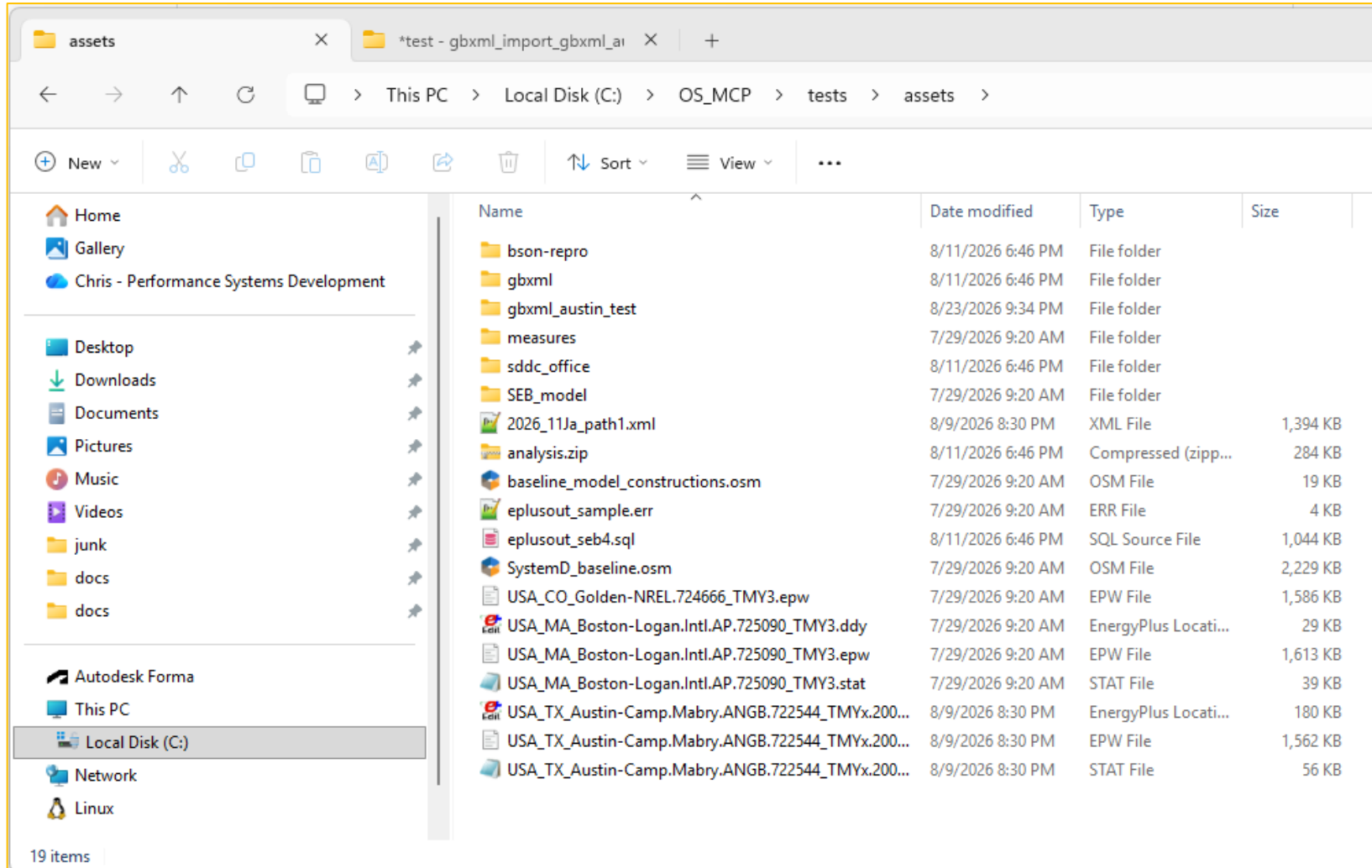
voidtools



**AI Agent
MCP Client**



Tool: gbXML import: ①



The OS-MCP
mounts to
this directory
(configurable)

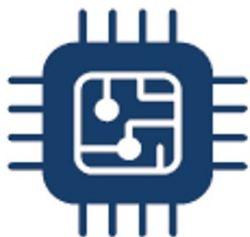
Tool: Repair and Validate gbXML: ②

Prompt

I just imported a gbXML file. Check the geometry for overlapping surfaces and spaces that aren't fully enclosed before I do anything else with it.



AI Agent MCP Client



Key Tools Used

Tool	Purpose
<code>import_gbxml</code>	Translate gbXML -> OSM (Example 21)
<code>repair_and_validate_gbxml_geometry</code>	Fix cross-space shared walls (<code>match_surfaces()</code> , internal) and re-sync desynchronized surface pairs, then report same-space overlaps, non-enclosed spaces and missing ground connections in one call
<code>repair_missing_roof_ceiling</code>	Follow-up fix for non-enclosed spaces that have a Floor but no RoofCeiling at all (see "Follow-up" below)
<code>merge_coplanar_sliver_surfaces</code>	Follow-up fix for non-enclosed spaces fragmented into many same-space coplanar pieces (see "Follow-up" below)
<code>weld_coincident_vertices</code>	Follow-up fix for non-enclosed spaces with sub-centimeter corner gaps between non-coplanar surfaces (see "Follow-up" below)
<code>patch_missing_surfaces</code>	Follow-up fix for non-enclosed spaces missing a surface (usually a partition wall) outright (see "Follow-up" below)
<code>trim_overlapping_surfaces</code>	Follow-up fix for non-enclosed spaces with a genuine same-space surface overlap (see "Follow-up" below)
<code>get_surface_details</code>	Only needed afterward, for a specific flagged surface pair, if the summary isn't enough to act on

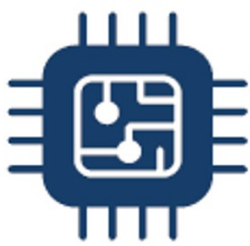
Tool: Attribute SpaceTypes to Spaces ③

Prompt

This building is mostly offices with a few conference rooms and corridors. Attribute space types to every conditioned space.



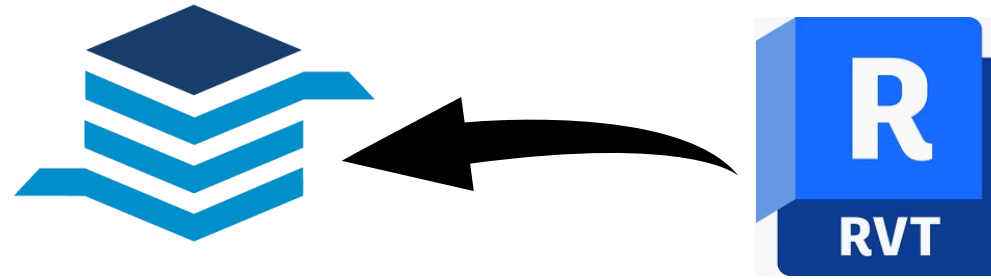
**AI Agent
MCP Client**



Key Tools Used

Tool	Purpose
<code>assign_space_type_simple</code>	One combo, every conditioned space, one call
<code>start_space_type_wizard</code>	Scan conditioned spaces, list available templates
<code>choose_space_type_templates</code>	Narrow to templates, list their building types
<code>choose_space_type_building_types</code>	Narrow to building types, show the first table page
<code>get_space_type_wizard_status</code>	Progress + remaining table, any time
<code>assign_space_type_batch</code>	Assign one combo to a batch of space indices
<code>finish_space_type_wizard</code>	Save + end, once every space is assigned
<code>cancel_space_type_wizard</code>	Abandon wizard tracking (does not undo assignments)

What can the OS MCP do?



01

Model Lifecycle

- Create
- Load
- Inspect
- Edit
- Save

02

Create and Edit Model Content

- Geometry
- Zones / Spaces
- Loads,
- Schedules,
- Constructions
- Materials

03

Add / Configure HVAC / SHW Systems

- “Standard” systems
- **ASHRAE Baselines**
- Customize Settings
- Customize Controls

04

Author NEW OS Measures

- Create **Custom OS Measures**, where OpenStudio’s custom analysis power lives.
- Python EMS / EnergyPlus Python Plugin authoring of custom controls, demand response, and custom metrics from natural language.
- Fallback only if measure not in NLR BCL bcl.nlr.gov

05

Scenario Management and Visualization

- Interactive 3D geometry, overlays and data can be created, ., including HTML reports.
- **Custom results dashboards** can be created

Use LLMs to call to structured software tools (not just chat) to directly manipulate real engineering models

OS-MCP: Scenario Management

Using my current model, I want to ...

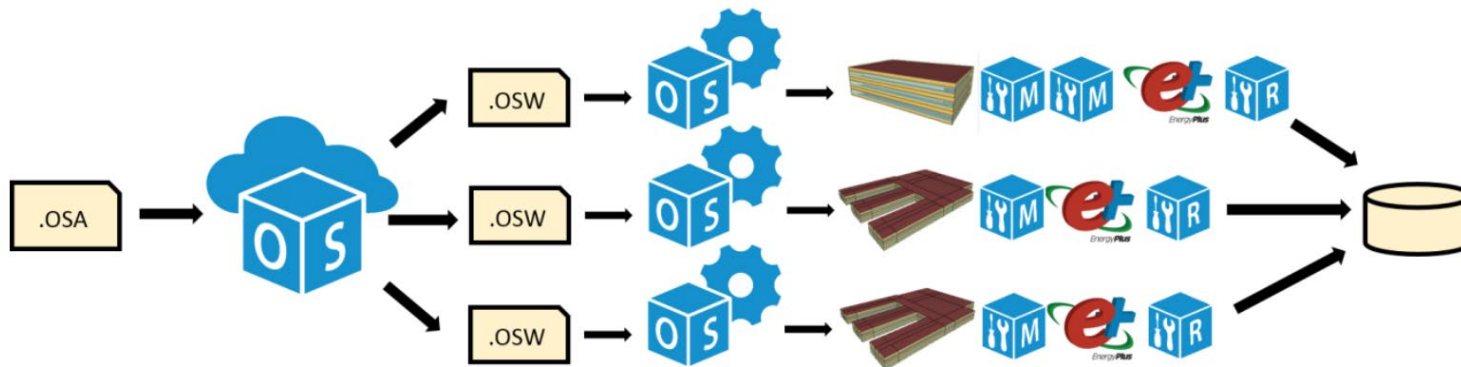


Examine the **Operational Energy** impact of the following sets of changes...

Examine the **Operational Carbon** impact from the following sets of changes...

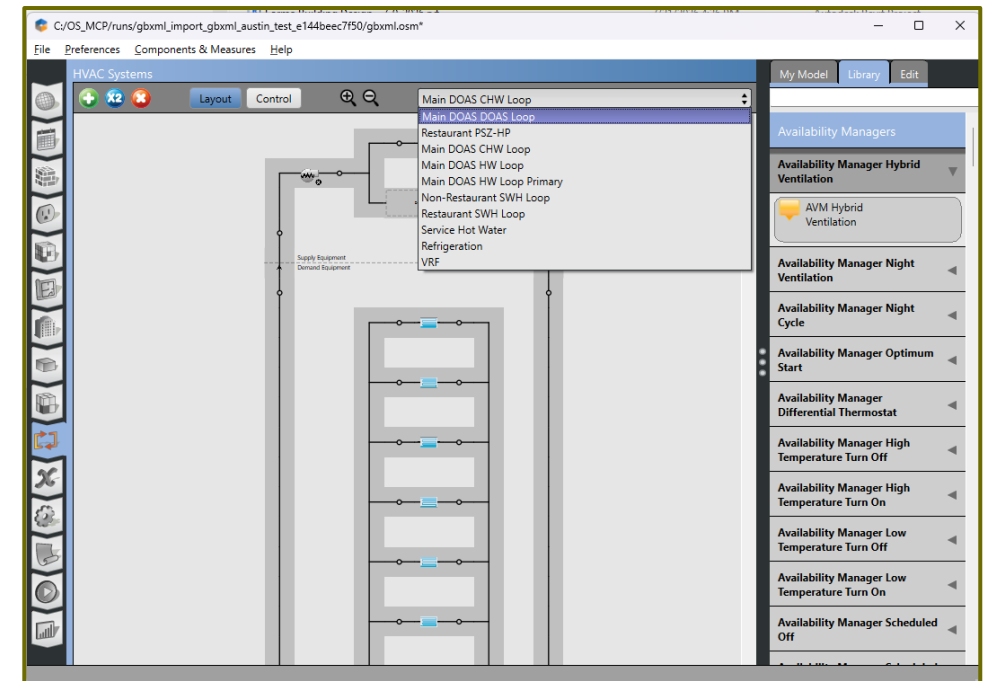
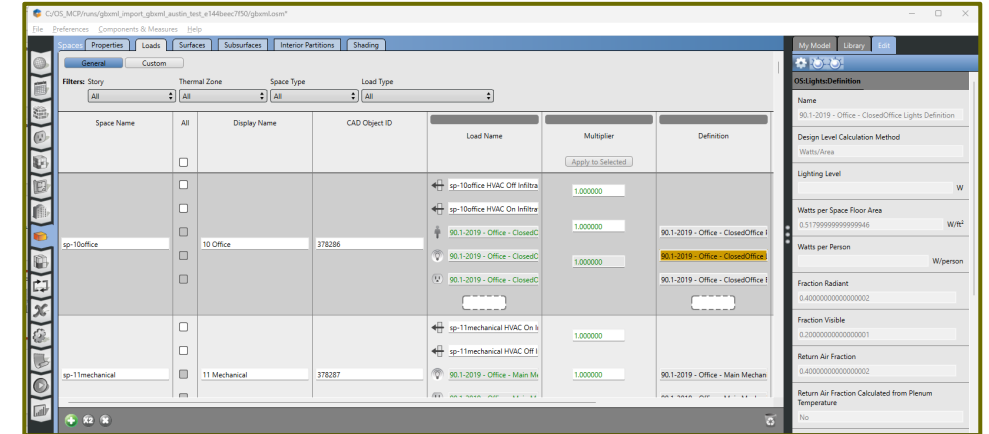
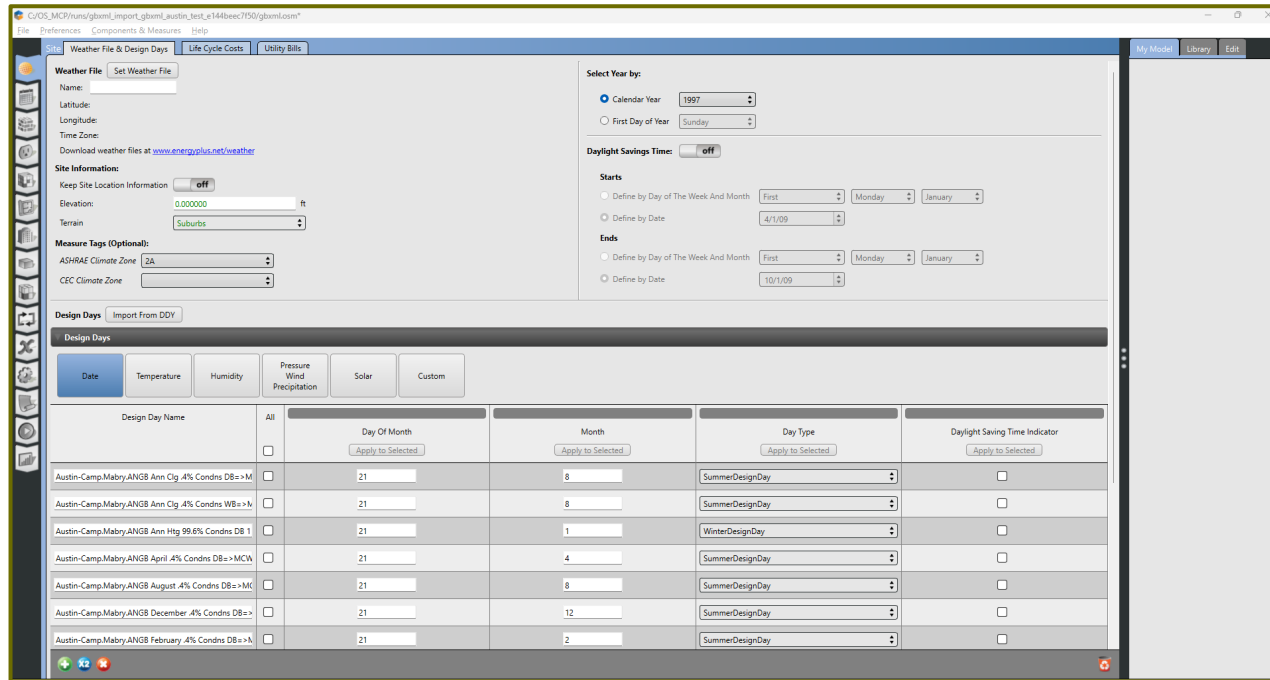
Examine the **Equipment Sizing** impact from the following sets of changes...

Examine the **Operating Cost** impact from the following tariff structures...



ADVANCED FEATURE
Connections to OS-
Server **are supported**
by the OS-MCP

OpenStudio Application



Model Inspection

<https://github.com/openstudiocoalition/openstudioapplication>

Revit export + OS-MCP Import



OS-MCP LIVE Demo (Time Permitting)



Concept Demo: Custom HVAC / Custom SHW

I want the conditioned spaces to be served by a 4 pipe fan coil system that uses a natural gas boiler and an air cooled chiller.

Lets change the `gas_boiler_efficiency` to 94%, the `fcu_fan_pressure_rise_in_h2o` to 1.2, and the `gas_boiler_primary_only_loop_pump_head_ft_h2o` to 75

+  Manual ▾

Sonnet 5 High ▾  

Volume Fan coils with air-cooled chiller with gas boiler and Variable Volume Fan coils with air-cooled chiller with gas boiler. Your description didn't specify which — do you want constant-volume or variable-volume fan coil fans?

A few things worth flagging: this includes zones like 403 EQUIPMENT and 405 MECHANICAL YARD, which came through the gbXML export as conditioned (they had heating/cooling setpoints), so I included them under "conditioned spaces" — let me know if you'd rather exclude utility/mechanical zones like that from active HVAC conditioning. No economizer auto-assignment was needed here since fan coil zone-equipment systems don't carry that requirement (only central air-loop systems do).

Does this look right, or does anything need to change?

everything looks good

Ran 5 commands

Locked in: **Variable Volume Fan coils with air-cooled chiller with gas boiler**, serving all 25 conditioned zones, standard `90.1-2004`.

Here are the custom properties available for this system (all optional — skip any and it falls back to the measure's built-in default):

Presented files

Done — one variable-volume fan coil system with an air-cooled chiller and gas boiler, serving all 25 conditioned zones under ASHRAE 90.1-2004, with your three custom overrides applied. It's validated and delivered above. Let me know if you'd like it saved back to your Autodesk project folder as well.

</>

Insight settings
Code · JSON

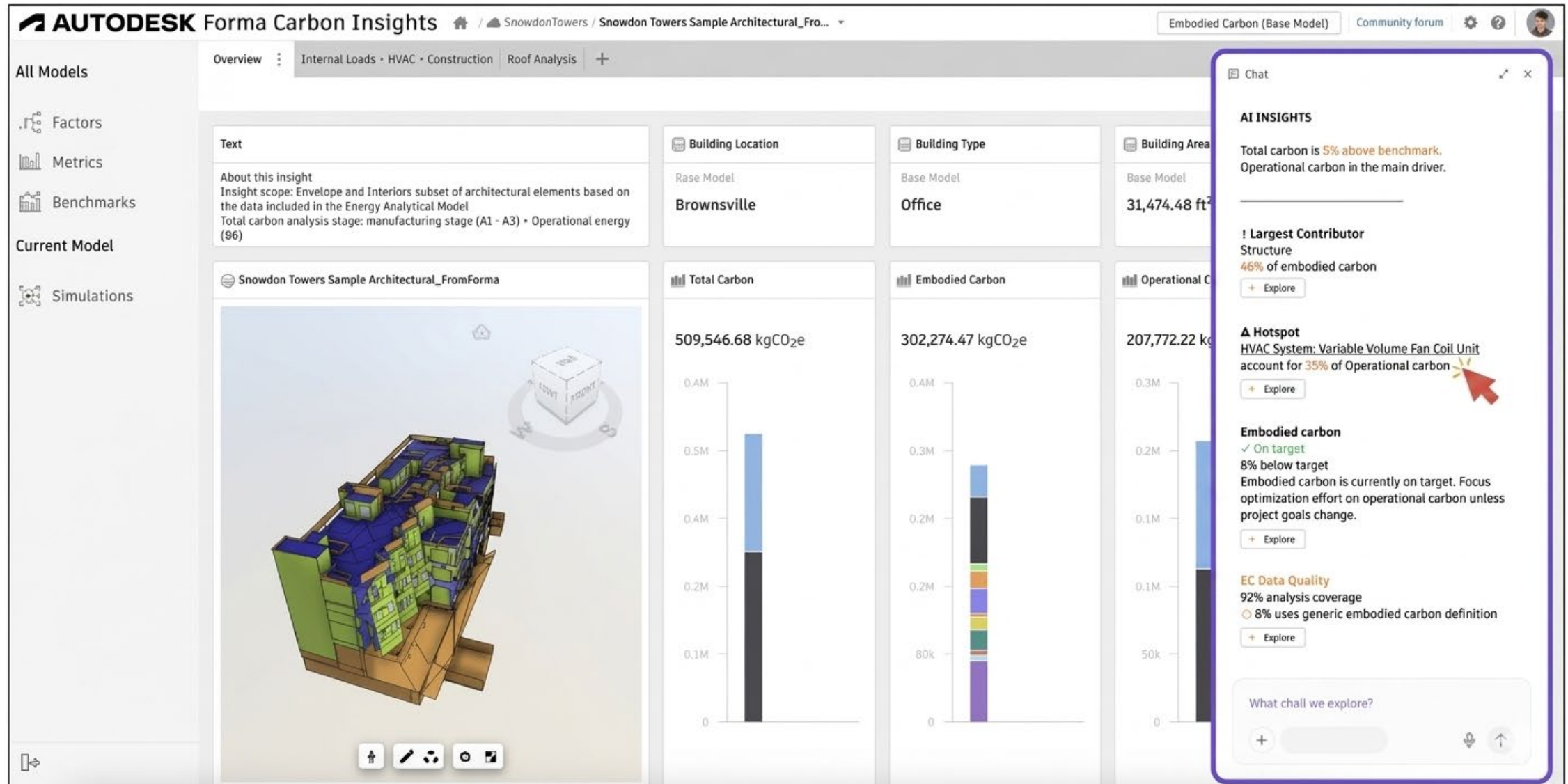
 Download and open ▾



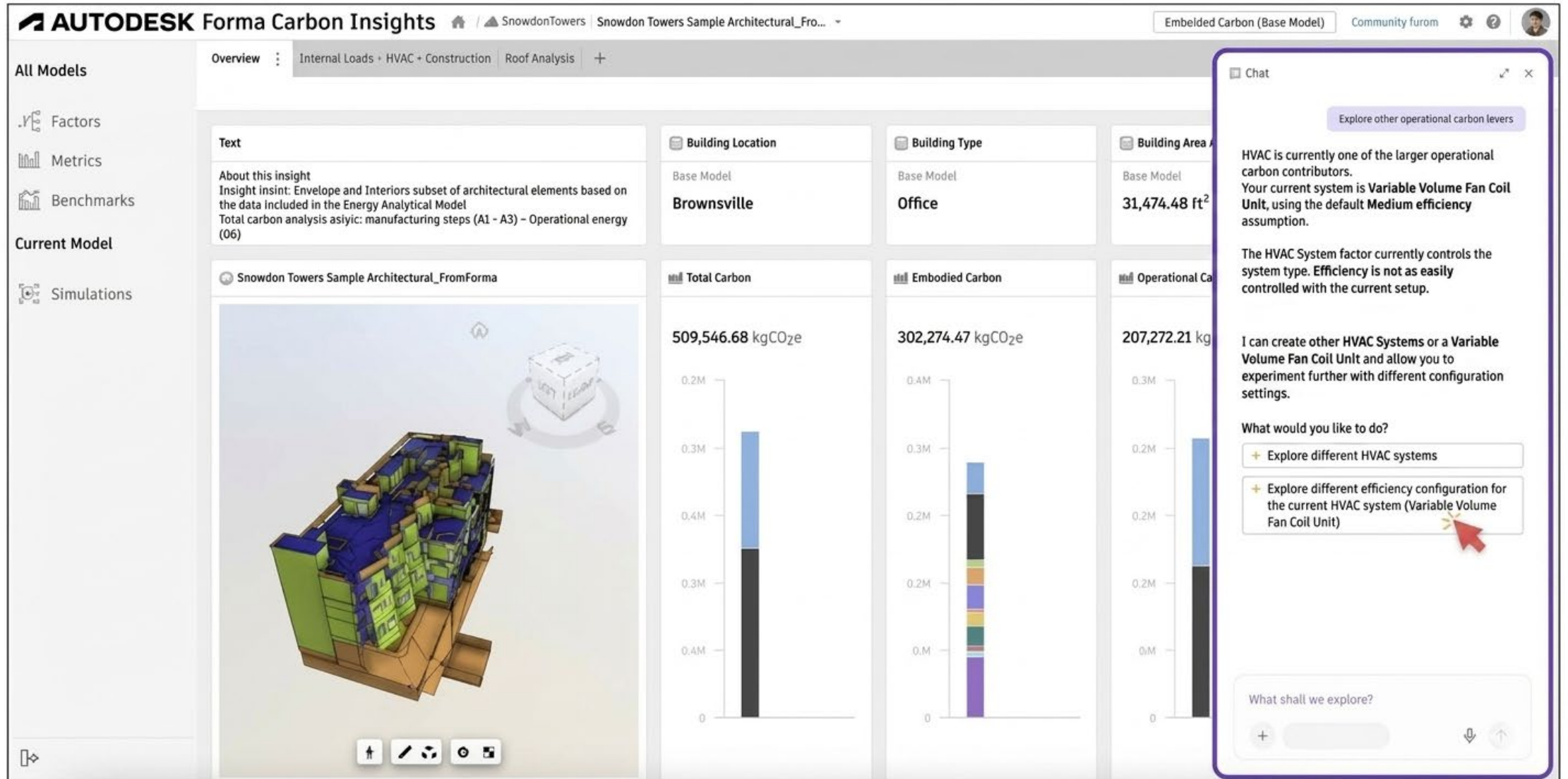
Gas boiler — plant operation & pumping

- `gas_boiler_plant_operation_scheme` — one of `SequentialLoad`, `UniformLoad`, `Optimal`
- `gas_boiler_primary_only_loop_pump_head_ft_h2o` — pump head, ft H₂O (5-500), used if primary-only

Concept Demo: Custom HVAC / Custom SHW



Concept Demo: Custom HVAC / Custom SHW



Leveraging **Connected Workflows** to Shape the Future



Forma Building Design

Forma Carbon Insights



NatLabRockies/openstudio-mcp

Overview of Related Sessions

Class Reference	Title	Type	Speakers
2634	Shaping Low-Carbon Buildings from the Start: A Hands-On Lab with Forma, Revit and Forma Carbon Insights	Hands-on lab	Giuseppe Ardito, Corina Marinescu, Justin Taylor, Ellis Herman
2018	Totally Carbon: Customizing Forma Carbon Insights to Your Project Needs	30 min Technical Deep Dive	Dattner – Dana Moore, Annabelle Swain
1983	From Massing to Metrics: Integrating Autodesk Forma and Autodesk Insight for Outcome-Based Design	Class	GHD (Australia) - Micheal Sturm – Jessica Carlson
2841	Forma to Energy: AI-Powered Workflows for Carbon Insight	60 mins - Class	PSD: Chris Ballbach, Dan Stine
3571	Connected Carbon Data for Construction: API-Powered Decisions at Scale	30 min Strategy Presentation	TBI (Netherlands) – Marta Bouchard



Conclusions And Q&A

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[AS2841 Autodesk Forma to Energy AI-Powered Workflows for Carbon Insight](#)

The logo for AU 2026 features a stylized 'A' icon composed of two overlapping geometric shapes: a dark gray trapezoid on the left and a white trapezoid on the right. To the right of this icon, the text 'AU2026' is displayed in a large, bold, black sans-serif font.

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THE DESIGN & MAKE CONFERENCE

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